NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

Low-altitude aircraft mainline

Group Low-Altitude Aircraft Platform

Status Closed

Period 2026-10-15 23:46 — 2026-11-13 17:00

Stage conclusion

Project report

R14 Low-altitude aircraft platform

Goal

For the new wind tunnel platform (owner: Xi'an University of Architecture and Technology; A aeronautical wind tunnel / B drone wind wall / C environmental chamber), propose requirements, protocols and acceptance criteria for drone wind-resistance and wind-disturbance estimation tests, and manage the comparison of the bidders' construction proposals (R14.1). The proposal materials do not constitute conclusions on equipment safety, airworthiness, compliance or construction approval.

Current status

Main line: protocol v1 and consultation draft v1 delivered (11-07); review by three independent reviewer models ruled (11 required changes); v1.1 delivered (11-07 evening); the lead checked that all 11 items were implemented. Whether consultation draft v1.1 goes out is waiting for the PI. Winter low-temperature E2 now has battery cards (11-07). Literature cards requested: aerodynamics and wind-resistance tests for the 7–25 kg

class, lithium battery low-temperature performance (literature agent, assigned 11-07). R14.1 weekend cleanup of to-dos that do not need the PI (assigned 11-07).

Decisions made

- 2026-10-20 Main line set as requirements and protocol for B wind-wall drone wind-resistance/wind-disturbance estimation tests; simulation deferred
- 2026-10-20 WAND disposition X2; method card added to the library
- 2026-10-26 Test aircraft spectrum split into two classes; option 4 compared first, not replaced
- 2026-11-03 Remaining items as recommended; U3DWind X4
- 2026-11-04 U3DWind point-by-point mean wind for the three Hong Kong cases not adopted (X2); ruling after review by six independent reviewer models

What we tried and results

Date	What was done	One-line result
2026-10-20	B-UAV test requirements v0	12+3 items, based on GB/T 38930, HB 8566, ISO 5110
2026-10-22	Test protocol v0, unified consultation draft v0	31 items; U-1…U-18 (10 hard, 7 to answer, 1 optional); written for a 3×3 m side wind wall, now outdated
2026-11-03—04	U3DWind X4 small test (two cluster jobs)	ε=0.210–0.212 > 0.15, window about 0.03 flow-through times → X2
2026-11-05	Winter low-temperature E2 (submitted directly by the PI; battery cards added 11-07: Li 2021, Shcherban 2024, Reid 2004)	Battery, density and wind-field stratification computed as three separate channels, X1

Open items and next steps

- Whether consultation draft v1.1 goes out (as supplementary technical requirements to Part B of the unified requirements document) → waiting for the PI.
- SAE J2777 and AIAA R-093 not obtained (library channel does not provide PDFs) → waiting for the PI: download via library account or purchase; marked "values to be verified" for now.
- Urban low-altitude wind reference data: public data must first be checked for spin-up and averaging time; winter stratified wind fields belong to R21/R16.5.

Literature basis

- PDF p.9: practical reference for LES spin-up and sampling length (U3DWind check)
- Wind-wall drone test cards (cited in the consultation draft and protocol)

Sub-projects

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Notes from the PI

Key figures